Windows 10 IoT Enterprise

servicing manual

Servicing manual for PCs with Windows 10 IoT Enterprise

Preamble

Windows 10 IoT Enterprise is specialized edition of Windows 10 aimed at deployment in industry. Most prominent feature is long-term security support without function updates. This provides stable environment for CNC control system,

System storage (solid-state drive) is separated to two partitions C: and D:. Partition C: is for system and applications, partition D: is for user data. Partition C: can protected against changes with Universal Write Filter[1] (UWF). As such, it is necessary to follow instructions in order to make changes that would otherwise be removed on reset by UWF.

Configuration Details

User accounts

System contains following predefined user accounts:

- **admin** (Machine Administrator)
  - default password **admin**
  - system administrator account controlled by customer
- **operator** (Machine Operator)
  - no password
  - limited user for machine operation
- **servicing** (Service Support)
  - password set by manufactured, must not be changed
  - system administrator account for manufacturer

Software preinstalled by vendor

- IntervalZero RTX – Windows real-time extension, required for Cnc886
- 7-Zip – archive manager
- Notepad++ – advanced text editor

1 More information available at https://docs.microsoft.com/en-us/windows-hardware/customize/enterprise/unified-write-filter

- ImageGlass – simple image viewer
- SumatraPDF – simple PDF viewer
- TeamViewer – remote support tool

Source code for open source programs is included in the installation in C:\Install.

Additional software

Any additional software must be approved by machine vendor prior installation. No warranty for compatibility is provided without previous testing.

Support scripts

Folder C:\Program Files\Tools contains number of scripts for system maintenance.

- **commit_cnc886.ps1(cmd)**– save changed files in C:\Cnc886 folder when UWF is enabled.
- **set_network_private.ps1** – Switches network profile to Private tin order to enable file sharing services.
- **updates_install(_restart).ps1** – Search and install all Windows updates (and automatically reboot computer).
- **updates_cleanup(_restart).ps1** – Remove unnecessary files left by Windows Update (and automatically reboot computer).
- **uwf-disable.ps1(cmd)** – Disable Universal Write Filter.
- **uwf-enable.ps1(cmd)** – Enable Universal Write Filter.

Further configuration details

- Windows Updates are controlled by customer IT management.
- Windows Defender files are relocated to hidden folder D:\WinDefend in order to keep these files writable even with UWF enabled.
- Remote Desktop is enabled for administrator accounts.
- During system start-up prompt appears enabling to choose between “Windows 10 – RTX” or „Windows 10“. For regular operation “Windows 10 – RTX” must be selected. This option is default and will be automatically selected after 5 seconds.

Details regarding the control application Cnc886

- Program files and most of the configuration files are stored in C:\Cnc886.
- Additional configuration files are stored in D:\config.
- CNC programs are stored in D:\cncfiles.
    - This folder is by default shared with read/write access for Everyone.
- Other user data are stored on drive D:.

Basic computer operation

- Power on: computer is configured to power on automatically as soon as machine is powered up.
- Power off: Switch Cnc886 to Blocked state, terminate Cnc886, shut down main power switch (optionally you may first shut down Windows via Start>Power Icon>Shutdown)
- Reboot: Switch Cnc886 to Blocked state, terminate Cnc886, reboot Windows via Start>Power Icon>Restart

How to make changes – standard procedures with UWF enabled

Making changes requires administrative privileges. Either the system requests them automatically or you must run given command as an administrator. If you are logged in as Operator, you will be prompted with dialog offering to log in as *admin* or *servicing* user. If you are already logged in as administrative account, you will be prompted with confirmation dialog.

Making changes in Cnc886 folder

Following steps are necessary to preserve changed files in folder C:\Cnc886. Files on drive D: are always preserved.

1. Terminate Cnc886 application
2. Perform needed changes (replace files, modify .knf files, …)
3. Run script **C:\Program Files\Tools\commit_cnc886.ps1** (right click > Run with PowerShell).
4. Script will request administrative privileges.
5. Reboot the system once the script is finished.
6. Verify that changes are preserved.

Changing system configuration

Following steps are necessary to change Windows configuration (e. g. network settings, user profile modifications, …)

1. Disable UWF.
    - Run script **C:\Program Files\Tools\uwf_disable.ps1** (right click > Run with PowerShell).
    - System will reboot.
2. After reboot, the UWF filter is disabled and all changes from this point forward will be saved to disk.
3. Perform all required changes.
4. Reboot computer.

5. Enable UWF.
    - Run script **C:\Program Files\Tools\uwf_enable.ps1** (right click > Run with PowerShell).
    - System will reboot.
6. After reboot, the UWF filter is enabled and any changes will be discarded on reboot.

Updating Windows

Note: Windows Update require Internet connection. Update process can take a long time during which the machine should not be used.

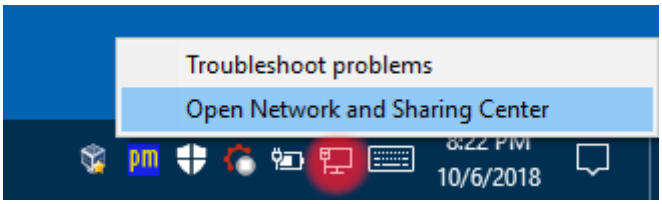
Before installing updates, check with machine vendor if updates have been tested for compatibility issues.

1. Disable UWF.
    - Run script **C:\Program Files\Tools\uwf_disable.ps1** (right click > Run with PowerShell).
    - Press ENTER to reboot system.
2. After reboot, the UWF filter is disabled and all changes from this point forward will be saved to disk.
3. Run script **C:\Program Files\Tools\updates_install(_restart).ps1** (right click > Run with PowerShell).
4. Once finished, reboot system and wait for updates to finish.
5. Repeat steps 3 and 4 as long as new updates show up. Repeat these steps at least twice.
6. Run script **C:\Program Files\Tools\updates_cleanup(_restart).ps1** (right click > Run with PowerShell).
7. Once finished, reboot system..
8. Enable UWF.
    - Run script **C:\Program Files\Tools\uwf_enable.ps1** (right click > Run with PowerShell).
    - Press ENTER to reboot system.
9. After reboot, the UWF filter is enabled and any changes will be discarded on reboot.

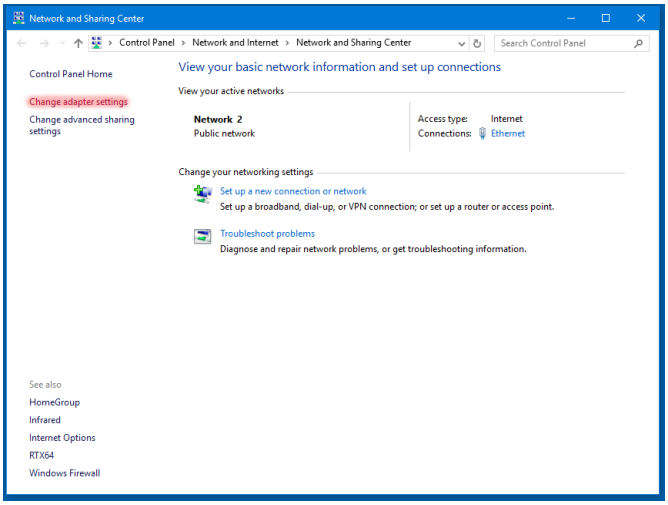
Configuring network address

Configuration of network requires disabled UWF, see chapter “Changing system configuration“.

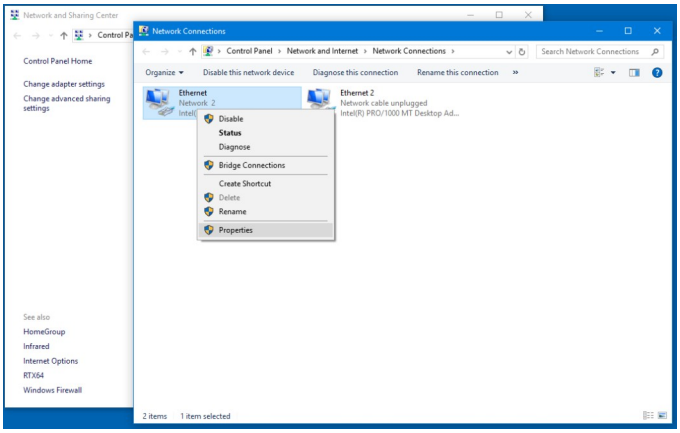
1. Open Network and Sharing Center by right-click on network icon.



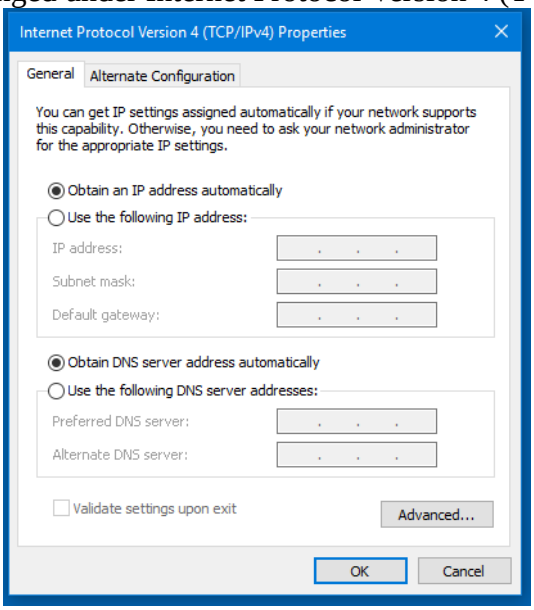
2. Open “Change adapter settings”



3. Open Properties of chosen adapter. Administrator login is required.



4. IP address can be changed under Internet Protocol Version 4 (TCP/IPv4)



5. Confirm all changes, reboot computer to verify it, and re-enable UWF.

Backup

- Critical files are Cnc886 configuration files. We recommend periodical backup of whole C:\Cnc886 folder.

- Backup regime for CNC files (stored in D:\cncfiles) is up to customer requirements.
- Additional data on D: drive (with the exception of D:\WinDefend folder) shall be backed up as necessary.
- Complete system backup can be performed by any available disk cloning tool. Tested tools include CloneZilla, Windows DISM disk capture using Windows PE.

Recommendations

- Restart computer periodically. Do not leave the system running for long periods. UWF consumes RAM and long periods of operation can consume all available RAM rendering the system unstable or unusable.

Known issues

- During boot, system keeps performing disk check on drive C:.
  - Skip disk check by pressing key when prompted.
  - Disable UWF and restart system.
  - Let the disk check run to completion.
  - Re-enable UWF and continue using the system.